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MID SEM REPORT

CHEMKIN-Based Investigation of the Effects of Compression Ratio and EGR on NH_3, NO_x and N_2O Emissions in an Ammonia–Diesel Dual-Fuel Compression-Ignition Engine

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1. Introduction

Ammonia is being investigated as a low-carbon fuel for internal-combustion engines because it contains no carbon. Its high auto-ignition temperature and slow combustion chemistry make pilot-diesel ignition attractive for compression-ignition operation. However, ammonia–diesel dual-fuel combustion introduces a coupled nitrogen-emission problem involving unburned NH_3 (NH_3 slip), NO_x and N_2O. Their formation depends on temperature, equivalence ratio, oxygen availability, residence time and radical chemistry.

The nine papers reviewed in this report collectively cover ammonia energy ratio and load, EGR, EGR mechanisms, injection strategy, compression ratio, spatial pollutant formation and detailed chemical kinetics. The literature is synthesized below to establish the scientific basis for the proposed CR × EGR CHEMKIN study.

2. Detailed Literature Review (9 Papers)

Chen et al. (2024) – AER and load

Energy 302, 131860. DOI: 10.1016/j.energy.2024.131860.

Chen et al. studied AER values of 15, 30, 45 and 60% at different loads. Increasing ammonia lowered combustion temperature, cylinder temperature and pressure, while the heat-release behavior changed with ignition delay. At full load, the D40A60 case showed large reductions in CO_2 and NO compared with D100, but N_2O increased because of partial ammonia oxidation. The study therefore demonstrates the fundamental AER–load trade-off: higher ammonia substitution can reduce carbon emissions and some NO, while incomplete ammonia oxidation can increase N_2O. Higher load also improves ammonia utilization.

Research gap: AER was investigated only over a limited range. N2O formation mechanism is not deeply resolved. The paper Reports that N2O increase with AER but the important unanswered question is which reaction is responsible for the increase.

Nie et al. (2024) – EGR under different altitudes

Energy 313, 133953. DOI: 10.1016/j.energy.2024.133953.

Using a 3.0-L four-cylinder ADDF engine at 1400 rpm and 50% load, Nie et al. investigated EGR at 0, 1000 and 2000 m altitude. Increasing EGR reduced in-cylinder pressure and heat-release rate and prolonged combustion. NO_x and NH_3 were reduced, with stronger reductions at higher altitude. Response-surface methodology was then used to optimize EGR strategy. The paper establishes that EGR effectiveness depends on the operating environment and is not simply a universal percentage reduction.

Research gap: N2O is not the central idea. No detailed chemical mechanism analysis. There is no simultaneous three emission optimization.

Jin et al. (2024) - EGR heat-capacity, dilution and chemical effects (2024)

Journal of the Energy Institute 117, 101778. DOI: 10.1016/j.joei.2024.101778.

This study separated the heat-capacity, dilution and chemical effects of EGR in an ADDF engine at high load and 80% AER. The heat-capacity effect increases mixture heat capacity and lowers average temperature, suppressing thermal and fuel NO_x but tending to favor N_2O-related low-temperature chemistry. The dilution effect lowers oxygen availability and heat-release intensity and was reported to reduce NO_x and N_2O. The chemical effect changes reactive groups and unburned components and can increase heat release and NO_x. The combined EGR effect reduced N_2O and strongly reduced NO_x in the investigated condition. This paper gives the strongest physical explanation for why EGR can have competing effects.

Research gap: The work separated the heat-capacity, dilution and chemical effects of EGR effectively, but the analysis remained focused on selected operating conditions and EGR/SODI optimization. The paper emphasizes Nox and N2O mechanism, but its final optimization is not formulated as min (NH3, Nox, N2O).

Xing et al. (2025) — Literature Review

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DOI: 10.3390/en18184841.

Xing et al. (2025) numerically investigated the effects of compression ratio (CR) and diesel injection timing on combustion and emissions of an ammonia/diesel dual-fuel engine under heavy-load conditions. A 3-D CFD model was used with a detailed ammonia/diesel chemical mechanism. The study showed that increasing CR improved ammonia oxidation and thermal efficiency and generally reduced NH_3 slip, while the higher combustion temperature increased NO_x formation. N_2O showed a non-monotonic behaviour due to the competing formation and destruction reactions. The study identified the combined influence of CR and injection timing on combustion performance and nitrogen-containing emissions.

Research gap: Research gap: The study used a fixed AER. The simulations were performed at 40% AER. So the result he produced may be not valid for higher AGR. The injection timing is also depend on AER. Nox problem cannot be solved and N2o problem is also solved at higher CR. Advanced injection timing is difficult at higher because if done then the negative work done on the piston will be higher which reduced ITE.

Sun et al. (2025) – EGR and injection strategies

Energy 315, 134391. DOI: 10.1016/j.energy.2025.134391.

Sun et al. investigated EGR together with injection strategy. EGR effectively controlled NO_x, but the ammonia/diesel engine was less tolerant of high EGR. Injection timing/pressure optimization could compensate for part of the combustion deterioration at high EGR. This is important because EGR changes oxygen concentration, ignition behavior and heat release; injection strategy changes the resulting temperature and equivalence-ratio fields. The study supports a coupled-parameter approach.

Research gap: N2O is not the main focus here. There is no detailed chemical pathway analysis. There is no clear simultaneous optimization for all nitrogen pollutants.

Hoseinpour et al. (2024)

Hoseinpour et al. (2024) numerically investigated the effects of intake charge temperature (330–390 K) and EGR rate (0–20%) on combustion and emissions of an ammonia/diesel dual-fuel engine under high-load conditions using 3-D CONVERGE CFD. Different ammonia energy shares were also considered. The study found that increasing intake temperature and EGR can significantly influence nitrogen-containing emissions. In particular, NH_3 emissions decreased at higher intake temperatures, while appropriate EGR conditions reduced NO_2 emissions. The authors also developed regression/response-surface models to identify favourable operating combinations.

Research gap: he generally focuses on Nox and NH3 optimization by varying EGR and intake temperature and does not focuses on N2O. actually he found the optimum EGR temperature combination for all the three emission. But they did not demonstrate that conclusion in the publicly available abstract. There is no detailed chemical pathway explanation.

Nie et al. (2025) – Nitrogen-containing pollutants

Energy 324, 136003. DOI: 10.1016/j.energy.2025.136003.

Nie et al. used CONVERGE to investigate NH_3, NO, NO_2 and N_2O in a 3.0-L four-cylinder ammonia/diesel engine under different operating conditions. Increasing AER reduced high-temperature regions and NO formation while increasing unburned NH_3. NH_3 distribution correlated with temperature, high-temperature regions favored NO formation, and N_2O formed around regions associated with NO. The work is highly relevant because it links engine-out emissions with in-cylinder temperature and mixture conditions rather than treating exhaust concentrations as isolated values.

Research gap: The study provided detailed information on the formation characteristics of NH_3, NO, NO_2 and N_2O under changes in AER, speed, load and atmospheric pressure. The combined effect of compression ratio and EGR on these pollutants was not systematically studied.

Peng et al. (2026)

He investigated the control of NH_3 and NO_x emissions from an ammonia/diesel dual-fuel engine using CONVERGE, GT-POWER and Response Surface Methodology (RSM). They studied the effects of ammonia energy ratio (AER), main-injection timing and injection pressure at 1400 rpm and 50% load. Increasing AER increased unburned NH_3 but decreased NO_x, while advancing injection timing and increasing injection pressure reduced NH_3 but increased NO_x. RSM identified an operating condition of 32% AER, 2°CA ATDC injection timing and 140 MPa injection pressure that reduced both NH_3 and NO_x under the studied conditions..

Research gap: the main focus is only on NH3 slip and Nox and there is no focus on N2O which is highly impactful green house gas

Chen et al. (2024) – Engine-out nitro-compounds

Fuel 362, 130740. DOI: 10.1016/j.fuel.2023.130740.

Chen et al. studied engine-out NH_3, NO, NO_2 and N_2O from a pilot-diesel-ignited ammonia/diesel engine. Unburned NH_3 increased with AER but decreased with load and speed. NO + NO_2 showed a nonlinear dependence on ammonia content and a trade-off with N_2O, while N_2O followed a parabolic-like trend with AER. The work demonstrates that nitrogen pollutants cannot be optimized independently and that injection strategy can affect NH_3 and N_2O.

Research gap: The study established important experimental relationships among ammonia energy ratio, engine operating conditions, NH_3, NO_x and N_2O emissions, and injection timing. He shows that advance SOI can reduce NH3 and N2O but increase Nox.

3. Cross-Paper Comparison

Paper	Year	System	Main variables	Method	Major finding	Main limitation
Chen	2024	NH_3/diesel CI	AER, load	Engine	AER lowers T/NO but can raise N_2O	No CR×EGR
Nie	2024	3-L ADDF	EGR, altitude	Experiment + RSM	EGR reduces NO_x/NH_3	No CR coupling
Jin	2024	ADDF	EGR, SODI	Experiment + numerical	Heat-capacity/dilution/chemical effects	No CR×EGR map
Xing	2025	NH3/diesel	CR, injection timing	3D CFD(Converge)	Higher CR ratios improve ITE and NH3 oxidation but increases Nox; N2O shows non monotonic behavior.	No EGR/ No CR/EGR kinetics
Sun	2025	ADDF	EGR, injection	Numerical/engine	High-EGR penalty can be mitigated	No CR
Hoseinpour	2024	NH_3/diesel	EGR, intake T, NH3 share	3D CFD + regression/ RSM	EGR and intake temperature influence NH_3 and NO_x emissions; higher temperature reduced NH	No CR+EGR; N2O not systematically optimized
Nie	2025	3-L ADDF	AER, speed, load, pressure	CONVERGE	Spatial NH_3/NO/N_2O trends	No CR×EGR kinetics
Peng	2026	NH_3/diesel	AER, injection timing, injection pressure	Converge, GT power, RSM	Optimized NH3 and Nox emissions	No N2O, no CR, EGR
Chen	2024	Pilot-diesel ADDF	AER, load/speed	Engine study	NH_3/NO_x/N_2O trade-off	No CR×EGR kinetics

4. Consolidated Research Gaps

1. A systematic CR × EGR investigation for pilot-diesel-ignited NH_3/diesel CI combustion is insufficiently established. Existing studies generally emphasize EGR, AER, load, injection parameters or CR separately or in selected combinations.

3. The coupled chemical effect of CR and EGR is not adequately explained. CR changes the compression thermodynamic state, while EGR changes heat capacity, dilution and radical chemistry.

4. Engine and CFD studies provide emission trends and spatial distributions, but a detailed CHEMKIN can identify the controlling reactions behind those trends.

5. The literature generally reports individual optima or parameter trends; an operating-window approach that balances all three target pollutants is still needed for the proposed application.

5. Research Objective

To numerically investigate the coupled effects of compression ratio and EGR rate on combustion and nitrogen-containing emissions in a pilot-diesel-ignited ammonia–diesel dual-fuel compression-ignition engine, with emphasis on identifying an operating window for simultaneous reduction of NH_3 slip, NO_x and N_2O using detailed chemical kinetics.

Specific objectives:

Select a suitable NH_3 + diesel-surrogate mechanism for CHEMKIN.

Represent compression/expansion using an engine-relevant cylinder-volume/crank-angle history.

Perform a controlled CR × EGR parametric matrix.

Evaluate NH_3, NO, NO_2 and N_2O together with temperature, pressure and heat release.

Use the simulation results and literature to interpret the observed emission trends.

Generate CR–EGR contour maps and identify a practical operating window.

6. Proposed CHEMKIN Methodology and Workflow

1. Define baseline speed, load, initial pressure/temperature, diesel pilot quantity/timing, AER and EGR composition.

2. Select and preprocess the detailed NH_3 + diesel-surrogate reaction mechanism.

3. Set the cylinder volume history; introduce CR through engine geometry rather than only a constant-volume assumption.

4. Represent EGR through physically justified dilution and oxygen-reduction effects, including CO_2/H_2O where appropriate.

5. Run and check a baseline case before the full sweep.

6. Run a CR × EGR matrix; for example, 5 CR levels × 5 EGR levels = 25 cases.

7. Extract temperature, pressure, heat-release behavior and $NH_3/NO/NO_2/N_2O$.

8. Create contour maps and identify low-emission regions.

9. Compare trends with the nine-paper literature base and define the final operating window.

7. Expected Outcomes

CR–EGR contour maps for NH_3 slip, NO_x and N_2O.

A chemically explained operating window rather than a single-pollutant optimum.

8. Conclusion

The nine studies establish that ammonia–diesel dual-fuel combustion has a coupled nitrogen-emission problem. AER and load affect ammonia utilization and temperature; EGR changes heat capacity, oxygen concentration and chemistry; injection changes combustion phasing; and CR changes the thermodynamic state before ignition. These findings support a systematic CR × EGR study in which NH_3, NO_x and N_2O are evaluated together.

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